

Adresse
- rue: string - ville: string - codePostal: string - pays: string
+ toString(): string + equals(other: Adresse): boolean

Creneau
- jour: JourSemaine - heureDebut: Time - heureFin: Time
+ getDuree(): number + chevauche(autre: Creneau): boolean + contient(heure: Time): boolean + toString(): string

Time
- heures: number - minutes: number
+ toString(): string // "HH:MM" + toMinutes(): number + compareTo(other: Time): number + addMinutes(minutes: number): Time

Periode
- dateDebut: Date - dateFin: Date
+ contient(date: Date): boolean + getDuree(): number // en jours + chevauche(autre: Periode): boolean

Equipement
- nom: string - quantite: number - estFonctionnel: boolean
+ toString(): string

ConfigurationEtablissement
- horairesParDefaut: HorairesParDefaut - joursTravailler: JourSemaine[] - dureeCoursStandard: number - pauseDejeuner: Creneau - recreations: Creneau[] - reglesSpecifiques: RegleSpecifique[]
+ getHoraires(jour: JourSemaine): HorairesJournee + valider(): boolean

HorairesParDefaut
- heureDebut: Time - heureFin: Time - pauseDejeunerDebut: Time - pauseDejeunerFin: Time

ParametresGeneration
- mode: ModeGeneration - prioriteEleves: number - prioriteEnseignants: number - prioriteSalles: number - flexibilite: NiveauFlexibilite - ponderationsContraintes: Map<string, number> - optionsAvancees: OptionsAvancees
+ valider(): boolean + toJSON(): any

StatistiquesEtablissement
- nombreClasses: number - nombreEnseignants: number - nombreEleves: number - nombreSalles: number - tauxOccupationSalles: number - chargeEnseignantsMoyenne: number
+ toJSON(): any

PreferencesUtilisateur
- notificationsEmail: boolean - notificationsSMS: boolean - notificationsInApp: boolean - langue: string - theme: string - heureDebutJournee: Time - heureFinJournee: Time
+ updatePreference(key: string, value: any): void

RapportQualite
- score: ScoreQualite - contraintesViolees: Contrainte[] - pointsForts: string[] - pointsAmeliorer: string[] - suggestions: string[]
+ genererPDF(): Buffer + toJSON(): any

Resolution
- description: string - impact: ImpactResolution - actions: Action[]
+ appliquer(): void + annuler(): void

ScoreQualite
- scoreGlobal: number - scoreContraintesDures: number - scoreContraintesSouples: number - detailsParCategorie: Map<string, number>
+ toString(): string + getPourcentage(): number